MOHAMED KABBA

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PROFESSIONAL SUMMARY

specializing in high-performance backend infrastructure and distributed systems. Expert in architecting scalable microservices using Python, Node.js, and Java. Proven track record handling 35k+ events/sec and 1.2M+ daily queries using Kafka, PostgreSQL, and Redis. Seeking Summer 2026 Software Engineering Internship.

EDUCATION

Bachelor of Science in Computer Science

University of the Potomac — *Washington, DC*

Expected Graduation: **February 2028**

- **GPA: 3.75/4.00 (Honors List)**
- Concentration: Software Engineering
- Relevant Coursework: Data Structures & Algorithms, Computer Organization & Programming, Object-Oriented Programming, Database Systems, Research and Statistical Analysis

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/Node.js, Java, Go, SQL

Backend & APIs: FastAPI, Spring Boot, NestJS, Django, REST, GraphQL, gRPC, WebSockets

Data & Storage: PostgreSQL, Redis, MongoDB, ElasticSearch, pgvector, SQLAlchemy, Prisma

Infrastructure: Apache Kafka, RabbitMQ, Docker, Kubernetes, AWS (EC2, Lambda, SQS), CI/CD

AI & Tools: LangChain, OpenAI/Gemini APIs, RAG, Git, Prometheus, Nginx

BACKEND PROJECTS

Event Streaming Analytics Pipeline

Technologies: Apache Kafka, Python, Kafka Streams, PostgreSQL, Docker, Prometheus

- Built a 3-broker **Apache Kafka** cluster handling **35k+ events/sec** with <100ms lag; implemented automated failover for zero data loss.
- Designed multi-consumer architecture for real-time analytics and ML pipelines using horizontal partitioning for high-throughput scaling.

Distributed Task Queue System

Technologies: Python, RabbitMQ, Redis, PostgreSQL, Docker, Celery

- Engineered a **Celery/RabbitMQ** task queue processing **10k+ jobs/min** with a 99.9% success rate and dynamic worker scaling.
- Developed robust error handling using **Dead Letter Queues (DLQ)** and exponential backoff, achieving sub-second latency for priority tasks.

Nexus — AI Knowledge Base API

Technologies: Python, FastAPI, PostgreSQL, pgvector, Redis, OpenAI

- Developed a **FastAPI** knowledge base processing **1.2M+ daily queries** using **pgvector** for semantic search and RAG workflows.
- Optimized performance to a **120ms average response time** via an intelligent Redis caching strategy (95% cache hit rate).